INCH-POUND

MIL-C-63461 (AR)
AMENDMENT 9
5 February 1993
SUPERSEDING
AMENDMENT 8
7 October 1991

MILITARY SPECIFICATION

CONTAINER, AMMUNITION, METAL,
FOR PROPELLING CHARGES AND CARTRIDGES

This amendment forms a part of Military Specification MIL-C-63461 (AR), dated 20 July 1981, and is approved for use by the U.S. Army Armament, Munitions, and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.1 Scope. Delete "and for packing cartridges".

2.1 Specifications, Military:

Delete "MIL-W-12332 - Welding Resistance, Spot, Seam and Production....."

Substitute "MIL-W-12332 - Welding Resistance, Spot, Seam and Projection....."

PAGE 2

2.1 Standards, Military:

Delete "MIL-STD-1188 - Commercial Packaging of Supplies and Equipment"

Add "MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes Functional Curves of the Continuous Sampling Plans"

2.1 Drawings:

Delete: "US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)" and Substitute: "US ARMY ARMAMENT RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER (ARDEC)"

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Add: "9349398 - Container, Ammo Metal: PA103"

2.2 Other publications

Add "ASTM-D3951 - Commercial Packaging:

Delete "ASTM-D412 - Tension Testing of Vulcanized Rubber, Standard Method of

3.2 Containers: After "(dwgs.) 9331196 through 9331205" add "and 9349398"

* 3.3.1 Tensile strength (if applicable). Add the following subparagraph:

"3.3.1.1 Tensile strength of body material. The tensile strength of the body material shall not be less than 47,000 pounds per square inch (psi) when tested as specified in 4.5.1".

3.4 Painted surface. Delete in its entirety and substitute:

"3.4 Painted surface. The painted surfaces of the container shall comply with the requirements of TT-C-490. However, the film thickness and salt spray requirements of TT-C-490 shall not apply to the interior surfaces and interior welded spots and sharp corners of the exterior surface of the container."

3.5 Cover assembly. Delete in its entirety and substitute:

"3.5 Cover assembly. Upon assembly of the cover assembly to the body assembly (body weldment), the cover assembly shall freely engage and rotate within the body assembly. As the cover assembly is rotated with the "Y" handle in the open position, the "Y" handle shaft ends shall freely engage into the respective bayonet slots of the rim. When the "Y" handle is rotated toward the closed position, the "Y" handle shall push the spring clip out of the way and snap into place (closed position) and the spring clip shall return to its original position locking the "Y" handle. When the "Y" handle is returned to the open position it shall push the spring clip out of the way permitting a 180 Degree rotation of the "Y" handle to the open position with the spring clip returning to its normal position. Rotation of the cover assembly will permit disengagement from the body assembly."

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Add new paragraphs as follows:

"3.8 Air pressure test.

3.8.1 Air pressure of unpainted cover assembly. The unpainted cover assembly shall withstand an internal air pressure of three pounds per square inch (psi) minimum (min), for five (5) seconds under water, without apparent leakage occurring when tested as specified in 4.5.8.1.

3.8.2 Air pressure of unpainted body assembly. The unpainted body assembly shall withstand an internal air pressure of three psi min., for five (5) seconds under water, without apparent leakage occurring when tested as specified in 4.5.8.1.

3.8.3 Air pressure of painted container assembly. The sealed container shall withstand the internal air pressure as specified in the applicable container assembly drawing when tested as specified in 4.5.8.2. See 6.5 for a definition of the flow rate method."

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Table I First Article Inspection. Under "Body assembly (weldment) unpainted: add:

"d) Weld strength of stacking plug to body (see Note) 5 (samples) 3.2 4.5.10/Dwg. 9331684"

PAGE 6

Table I First Article Inspection.

1. Under "Gasket", Hardness: Delete "Dwg 9331685" and substitute "Dwg 12910349".

2. Under "Gasket", Delete "c) Tensile stress (see Note) 3 3.2 4.5.6/Dwg 9331685"

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3. Add the following:

"Cover weldment.

a) Examination for defects 10 (samples) 3.2 4.4.2.11
b) Weld strength (of shaft ass'y to cover) 3 (samples) 3.2 4.5.9/Dwg 9331692"

PAGE 8

4.4.2.1 Clamp. Delete Major Defects 102, 103, and 104 and substitute:

"Major

102 True position of holes with datums -A-, -B- and -C- 0.40% 3.2 Gage

103 True position of holes with datum -A- 0.40% 3.2 Gage"

4.4.2.2 Spring. Delete Major Defects 102, 103, and 104 and substitute:

"Major

102 True position of holes with datums -A-, -B- and -C- 0.40% 3.2 Gage

103 True position of holes with datum -A- 0.40% 3.2 Gage"

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4.4.2.3 Support, shaft. Delete Major Defects 104, 105, 107, 108, 111, 112 and 113 and substitute:

"Major

* 111 True position of rivet holes with datums -A-, -B- and -D- 0.40% 3.2 Gage

* 116 Height (1.27) of spring tab 0.40% 3.2 Gage"

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4.4.2.5 Handle. Classification of Defects, Delete:

"103. Vertical location of small hole... .40% 3.2 Gage
104 Horizontal location of small hole... .40% 3.2 Gage
201. Distance from center of shaft hexhole to outer edge of tabs. .40% 3.2 Gage".

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* 4.4.2.6 Shaft. All new major defect as follows:

"108. Position of large diameter with respect to datum -A- 0.40% 3.2 Gage"

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4.4.2.8 Retainer. Major Defects, Delete:

"103. Wall thickness .40% 3.2 Gage
104 Inside depth .40% 3.2 Gage"

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4.4.2.9 Cover.

Major defect 106. After "Height of threads (min.)", add "when applicable. (See note)."

"Notes" section. Add following note: "Threads may be formed on cover assembly (instead of cover)."

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4.4.2.10 Gasket.

1. Drawing number block. Delete "9331685", substitute "12910349"

2. Major defect 107. (Misalignment of joint). Delete in its entirety.

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4.4.2.11 Cover weldment. Add minor defects as follows:

"202. Distance from centerline of weld to edge (5 places) .65% 3.2 Gage
203. Distance from centerline of weld to edge (1 place) .65% 3.2 Gage"

PAGE 19

4.4.2.12 Cover assembly, prior to painting. Add major defects as follows:

"104. Distance from top of rod (in closed position) to inside surface of the gasket retainer section of cover (.725% ± .020) 0.40% 3.2 Gage

105. Height of threads (min.), when applicable (see note). 0.40% 3.2 Gage"

"Notes" section. Add following note "Threads may be formed on cover (instead of cover assembly)."

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4.4.2.14 Rim. Major defects 102 and 104. In each of these defects, change "hole" to "slot".

Major defects 103 and 105. Delete in their entirety and substitute:

"103. Diameter of small holes on side of rim 0.40% 3.2 Gage

105. Location of small holes on side of rim 0.40% 3.2 Gage"

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4.4.2.15 Ring.

Major defect 102. Change "large hole" to "large slot"

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Major defect 103. Delete in its entirety and substitute:

"103. Diameter of small holes on side of ring 0.40% 3.2 Gage"

Major defect 104. Delete in its entirety and substitute:

"104. Location of larger slot 0.40% 3.2 Gage"

Major defect 105. Location of small holes on side of ring 0.40% 3.2 Gage"

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4.4.2.18 Plug stacking. Delete all Major Defects and substitute:

"Major

101 Diameter of indexing projections 0.40% 3.2 Gage

102 Location of indexing projections 0.40% 3.2 Gage.

103 Length of plug 0.40% 3.2 Gage.

104 Width of plug 0.40% 3.2 Gage."

4.4.2.19 Body weldment. (Body Assembly), prior to painting.

Major defect 101. Delete in its entirety and substitute:

"101. True position of slots of ring(s), rim and holes of stacking plug 0.40% 3.2 Gage"

Major defect 102. Delete in its entirety.

Major defect 103. Delete in its entirety.

Add major defect as follows:

"106. .885 Location of bayonet slot of rim with respect to edge of body (Section A-A) 0.40% 3.2 Gage"

4.4.3.1 Tensile strength. Delete in its entirety and substitute the following:

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"4.4.3.1 Tensile strength. (see 3.3.1), Major Defect. Six (6) bodies shall be randomly selected for this test. From each body, three (3) specimens shall be selected at random for determination of body material tensile strength. Also, from each body, four (4) specimens approximately equally spaced along the axis shall be taken at right angles to the weld and the tensile strength of their welds shall be determined. Each of these values shall be compared with the lowest determinated value for the body material tensile strength for compliance with the requirement of 3.3.1. If one or more specimens (longitudinal) body weld or body material) fail to comply with 3.3.1 or 3.3.1.1, that sample body shall be classed defective. If two or more sample bodies are classed defective, the lot shall be rejected."

4.4.3.5 Deflection of load of spring. Delete in its entirety.

4.4.3.6 Tensile strength of gasket. Delete in its entirety.

4.4.3.7 Hardness of gasket. Delete "(see Dwg. 9331685)" and substitute "(see Dwg. 12910349)"

4.4.3.8.1 Air pressure of unpainted cover assembly. Delete "(see applicable container assembly dwg. (see 2.1)" and substitute "see 3.8.1".

4.4.3.8.2 Air pressure of unpainted body assembly. Delete "(see applicable container assembly dwg. (see 2.1)" and substitute "see 3.8.2".

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4.4.3.8.3 Air pressure of painted container assembly. Delete in its entirety and substitute:

"4.4.3.8.3 Air pressure of painted container assembly. (see applicable container assembly drawing (see 2.1)), Major defect. Container assemblies shall be selected in accordance MIL-STD-1235, continuous sampling plan CSP-1, using sample frequency Code D with a value of i of 135 consecutive units. The test shall be performed as specified in 4.5.8.2."

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Add new paragraph as follows:

"4.4.3.9 Weld strength of stacking plug to body. (See dwg. 9331684), Major Defect. Three (3) bodies shall be randomly selected for this test. From each body two (2) stacking plug welds shall be tested in accordance with 4.5.10. If one or both welds fail to comply with the requirement, the body shall be classed defective. If one or more bodies are classed defective, the lot shall be rejected."

* 4.5.2 Weld of rim to body. Add the following sentences:

"However, weld coupons shall not be taken. The test shall be performed on the rim-body weldment. The load, which is applied, may be either a tensile or compressive load in order to produce a weld shear."

4.5.5 Deflection load of spring. Delete in its entirety.

4.5.6 Tensile strength of gasket. Delete in its entirety.

4.5.8 Air pressure test. Delete in its entirety and substitute:

"4.5.8 Air pressure test.

4.5.8.1 Air pressure of unpainted cover assembly and unpainted body assembly (water immersion method). Subject the cover assembly and container body to the specified internal air pressure for the test time specified. For the cover assembly, apply the pressure when the cover assembly is assembled to a master body assembly (or its equivalent) equipped with a gasket. For the body assembly, apply the pressure when the body assembly is assembled to a master cover (or equivalent) and gasket. The master cover and the master body assembly shall be provided with satisfactory means of applying and maintaining the pressure during the test and for showing evidence of leakage. Observation shall be made for the presence of bubbles. No bubbles (other than surface bubbles) shall be permitted. A cover assembly or body assembly which does not meet the requirements, shall be classed defective.

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4.5.8.2 Air pressure of painted container assembly. Apply the internal air pressure, specified on the applicable container assembly drawing, to the container. Observation shall be made for a flow rate exceeding the requirements for the test time (as specified on the applicable drawing). The container which exceeds the requirements shall be classed defective. When cover assemblies or body weldments are procured separately, the contractor shall submit for Government approval, the test procedure that will be followed to meet the applicable test requirement on the applicable ammunition container drawing."

PAGE 30

Add new paragraph as follows:

"4.5.9 Weld strength of shaft assembly to cover. (First Article Inspection Only) - The strength of the weld shall be determined in accordance with ASTM-E8".

Add new paragraph as follows:

"4.5.10 Weld strength of stacking plug to body. The strength of the weld shall be determined in accordance with ASTM-E8".

5.2 Packing, second sentence:

Delete "MIL-STD-1188" and substitute "ASTM-D3951"

5.3 Marking, first sentence:

Delete "MIL-STD-1188" and substitute "ASTM-D3951"

6.2 Drawings: Delete in its entirety and substitute:

"6.2 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal, or US Army Armament Research and Development Command (ARRADCOM) drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC."

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6.3 Submission of inspection equipment for design approvals: Delete in its entirety and substitute:

"6.3 Submission of inspection equipment for design approvals. - See MIL-A-48078. Submit designs as required to: Commander, US Army Armament Research and Development Center, ATTN: SMCAR-QAR-I, Picatinny Arsenal, NJ 07806-5000.

6.4 Second sentence:

Delete "US Army Armament Research and Development Commmand, ATTN: DRDAR-QAR-Q, Dover, New Jersey 07801" and substitute "US Army Armament Research and Development and Engineering Center, ATTN: SMCAR-QAR-Q, Picatinny Arsenal, NJ 07806-5000".

Add new paragraph as follows:

"6.5 Flow rate method. The flow rate method is defined as a procedure to measure the passage of air or gas, at ambient temperature, from the container in a specified time."

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